

El-Beginner：具身智能入门练习

入门基础

- 会用ChatGPT/DeepSeek和Google
- 会用Linux
- 会用Git和GitHub
 - <https://learngitbranching.js.org/>

任务一：基于传统运动学的机械臂物体抓取

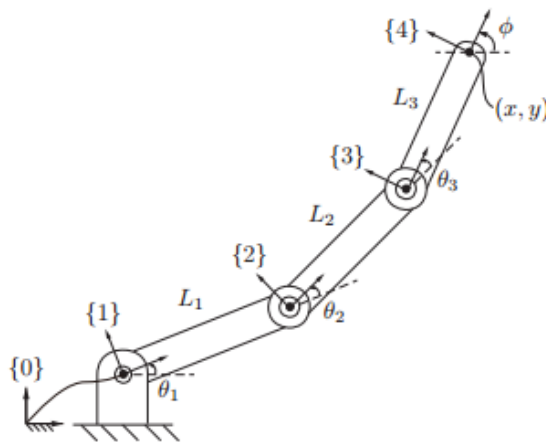


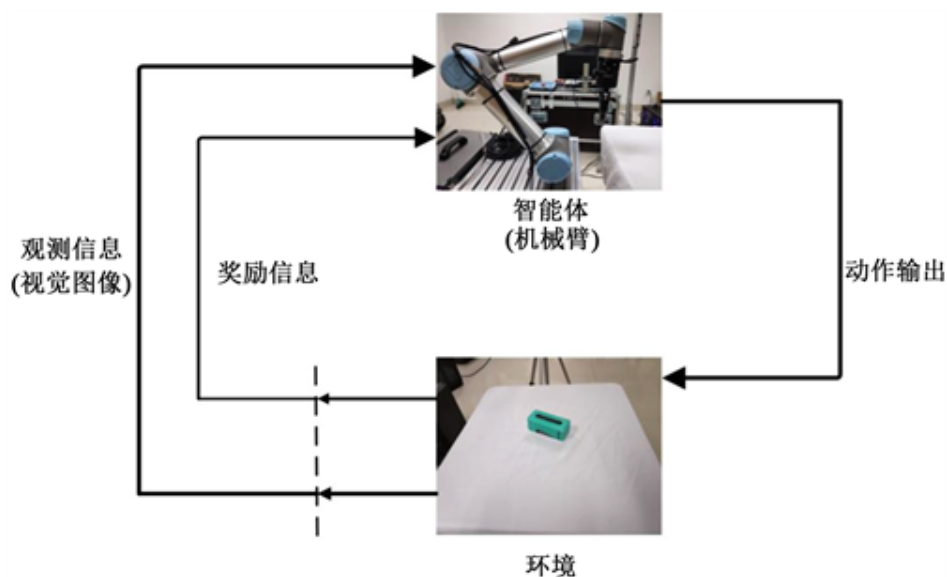
Figure 4.1: Forward kinematics of a 3R planar open chain. For each frame, the $\hat{x}$ and $\hat{y}$ axes are shown, and the $\hat{z}$ axes are parallel and out of the page.

学习传统机器人学中的基础知识，比如基础坐标变换、正逆运动学、动力学、控制理论等，在PyBullet/Mujoco仿真与实验室机械臂真机（可选, 同时学习机器人操作系统ROS）实现基于传统机器人运动控制的机械臂物体抓取；

备注：

- 参考书目
 - Introduction to Robotics: Mechanics and Control from Stanford
 - [Robotic Manipulation](#) from MIT
- PyBullet仿真链接： <https://github.com/bulletphysics/bullet3>
- Mujoco仿真链接： <https://mujoco.org/>
- 闯关游戏： <https://rcfs.ch/>

任务二：基于强化学习的机械臂物体抓取

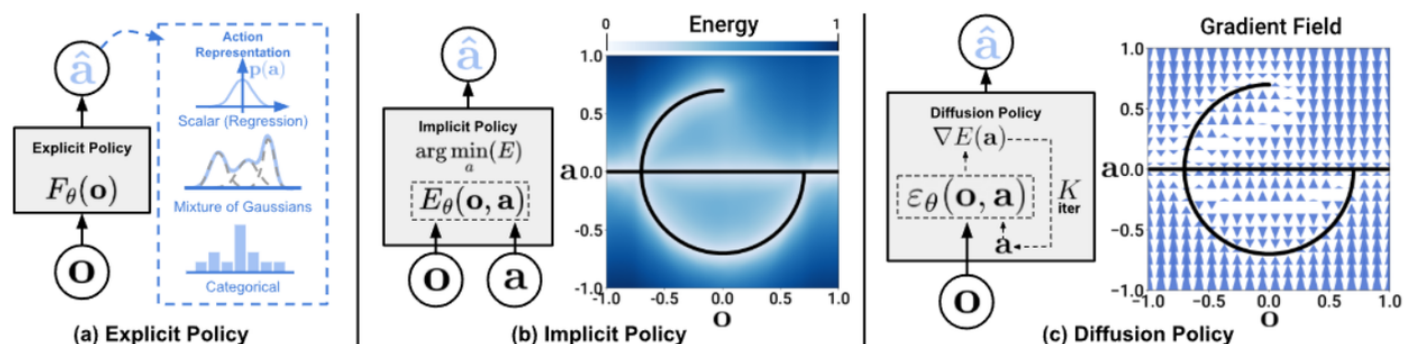


1. 学习强化学习基础知识，并在OpenAI Gym环境中选择几个感兴趣的tasks使用强化学习方法训练并测试成功率；
2. 在PyBullet/Mujoco中训练机械臂的物体抓取策略，并尝试在真机部署，体会Sim2Real的过程；

备注：

- 参考书目
 - 《神经网络与深度学习》相关章节
 - Introduction to Reinforcement Learning, 2nd & David Silver's UCL Course
 - UCB 285 Deep Reinforcement Learning
- OpenAI Gym链接：<https://gymnasium.farama.org/index.html>

任务三：基于模仿学习的机械臂物体抓取



1. 复现模仿学习经典baseline方法Diffusion Policy
 - a. <https://diffusion-policy.cs.columbia.edu>
2. 同时学习Huggingface机器人学习框架LeRobot
 - a. <https://github.com/huggingface/lerobot>

任务四：基于VLA大模型的机械臂物体抓取

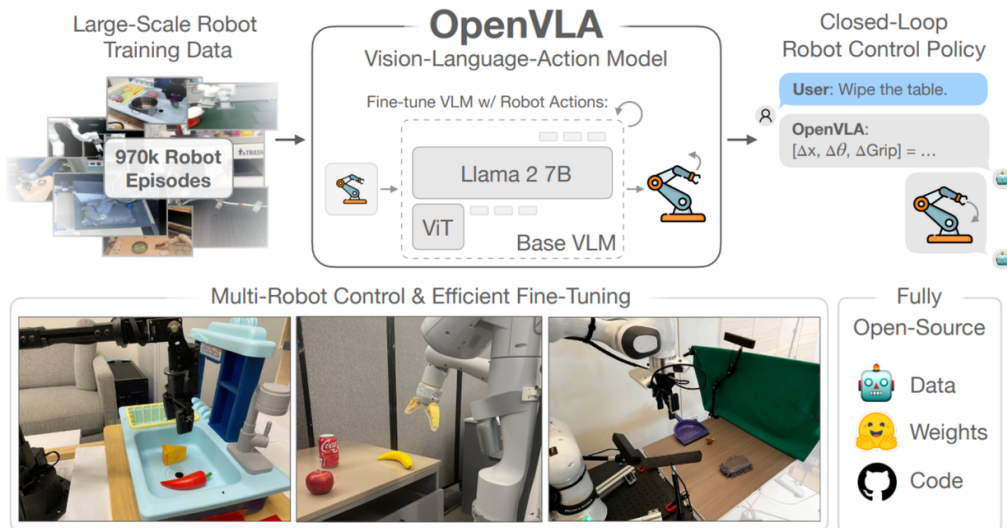


Figure 1: We present OpenVLA, a 7B-parameter open-source vision-language-action model (VLA), trained on 970k robot episodes from the Open X-Embodiment dataset [1]. OpenVLA sets a new state of the art for generalist robot manipulation policies. It supports controlling multiple robots out of the box and can be quickly adapted to new robot domains via parameter-efficient fine-tuning. The OpenVLA checkpoints and PyTorch training pipeline are fully open-source and models can be downloaded and fine-tuned from HuggingFace.

学习并利用现有的VLA大模型如OpenVLA/GR00t N1等, 探索利用现有的机械臂数据集如Open-X Embodiment训练专门的VLA机械臂模型

备注:

- OpenVLA: <https://github.com/openvla/openvla>
- Gr00t N1: <https://github.com/NVIDIA/Isaac-GR00T>
- Open-X Embodiment: <https://robotics-transformer-x.github.io/>
- Large Models: <https://stanford-cs336.github.io/spring2025/>

任务五：基于LLM/VLM大模型的任务规划

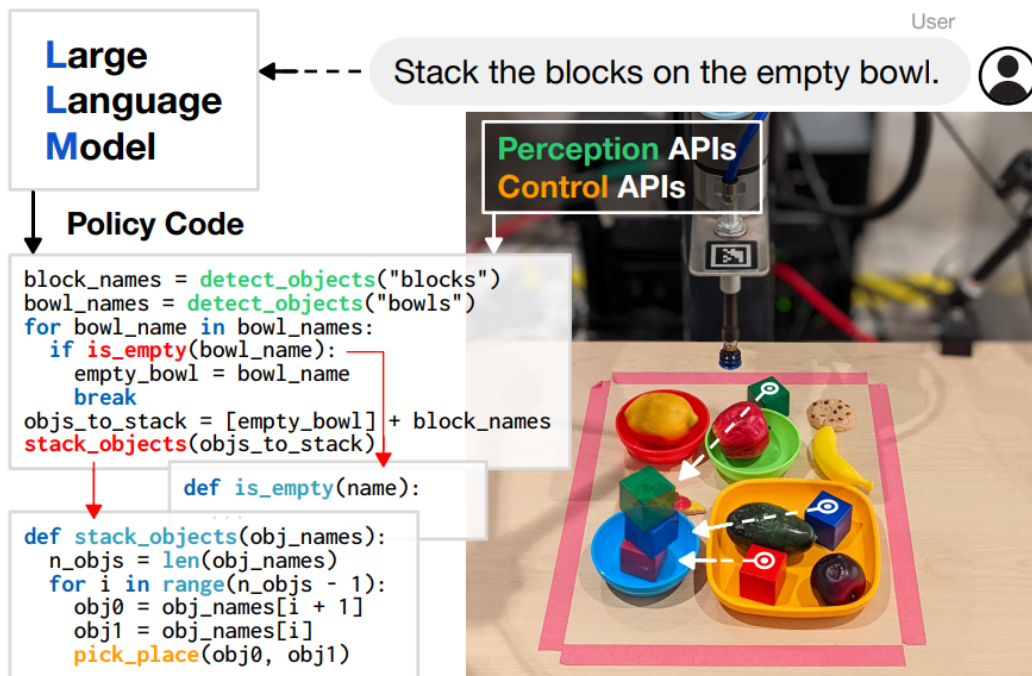


Fig. 1: Given examples (via few-shot prompting), robots can use code-writing large language models (LLMs) to translate natural language commands into robot policy code which process **perception** outputs, parameterize **control** primitives, recursively generate code for **undefined** functions, and generalize to new tasks.

1. 桌面级任务规划

- 参考论文 “Code as Policies: Language Model Programs for Embodied Control” (ICRA 2023 Outstanding Paper, <https://code-as-policies.github.io/>) 在PyBullet仿真与实验室机械臂真机（可选）实现基于视觉模块和大语言模型的机械臂物体抓取；
- Prompt现有的LLM/VLM来完成这个任务；
- Finetune现有的LLM/VLM来完成这个任务；

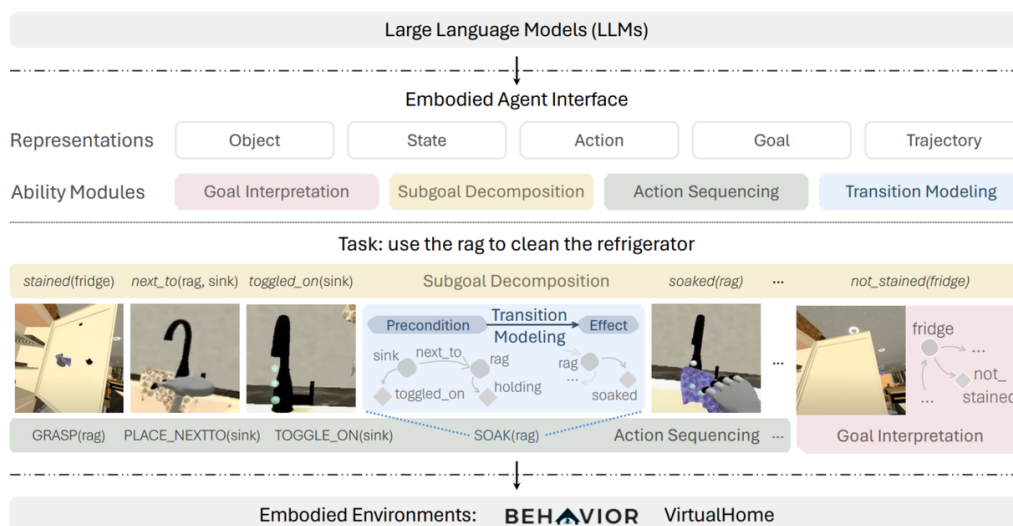


Figure 1: EMBODIED AGENT INTERFACE unifies a broad set of tasks involving both state and temporally extended goals and four LLM-based modules for decision-making.

2. 场景级任务规划

- 配置任意一个仿真环境，选择一个模型跑通baseline

- b. 设计icl或者cot方法，提高模型具身规划效果
- c. （可选）使用对应benchmark提供的训练集，sft模型对比效果

可选仿真环境/benchmark

- EAI <https://github.com/emodied-agent-interface/emodied-agent-interface>
- VAB-OmniGibson <https://github.com/THUDM/VisualAgentBench>

参考论文

- Embodied Agent Interface: Benchmarking LLMs for Embodied Decision Making <https://arxiv.org/pdf/2410.07166>
- VisualAgentBench: Towards Large Multimodal Models as Visual Foundation Agents <https://arxiv.org/pdf/2408.06327>
- LLM-Planner: Few-Shot Grounded Planning for Embodied Agents with Large Language Models <https://arxiv.org/abs/2212.04088>
- ReAct: Synergizing Reasoning and Acting in Language Models <https://arxiv.org/pdf/2210.03629>

任务六：基于强化学习的人形机器人运动控制



Figure 1: (a) OmniH2O enables teleoperating a full-size humanoid robot (Unitree H1) to complete tasks that require both high-precision manipulation and locomotion. (b) OmniH2O also enables full autonomy through visual input, controlled by GPT-4o or a policy learned from teleoperated demonstrations. **Videos:** see our website: <https://omni.human2humanoid.com>

复现OmniH2O: Universal and Dexterous Human-to-Humanoid Whole-Body Teleoperation and Learning (<https://omni.human2humanoid.com/>) 的人形机器人运动控制方法, 学习人形机器人基础知识, 仿真训练+Sim2Real的过程

1. 全身遥操作
2. 全身模仿学习

参考:

- Unitree Robotics GitHub: <https://github.com/unitreerobotics>
- Hover: <https://github.com/NVlabs/HOVER>
- [Underactuated Robotics](#) from MIT

前沿研究

如何做研究

- [GAMES003-图形视觉科研基本素养](#)

会议期刊

- 机器人: Science Robotics, RSS, CoRL, ICRA, IROS, RLC
- 机器学习: ICLR, NeurIPS, ICML
- 计算机视觉: CVPR, ICCV, ECCV
- 自然语言处理: ACL, EMNLP, COLM

研讨班

- CMU Robotics Institute Seminar: <https://www.youtube.com/@cmurobotics>
- MIT Robotics Seminar: <https://www.youtube.com/@MITRoboticsSeminar>
- MIT Embodied Intelligence Seminar:
<https://www.youtube.com/@mitembodiedintelligence8675>
- Stanford Seminar: <https://www.youtube.com/@stanfordonline>

